

The exceptional set of a  $c_0\ell_1$  family is countably separating  
Candidate full solution to the converse question after Corollary 4.10 of arXiv:2111.10670

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**Status.** Candidate full affirmative solution, subject to human verification. Combining the theorem below with source Corollary 4.10 gives an if-and-only-if characterization of weak-star-null families of type  $c_0\ell_1$  over a subset of a compact line.

## 1 Source and exact question

The source is V. dos Santos Ronchim and D. V. Tausk, *Extension of  $c_0(I)$ -valued operators on spaces of continuous functions on compact lines* [1], arXiv:2111.10670, published in *Studia Mathematica* 268 (2023), 259–289.

For a compact line  $K$ , a subset  $Q \subseteq K$ , and a weak-star-null family  $(F_i)_{i \in I}$  in  $\text{NBV}(K)$ , source Corollary 4.10 proves that

$$E := \{t \in Q : (F_i(t))_{i \in I} \notin c_0(I)\}$$

admitting a countably separating family implies that  $(F_i)_{i \in I}$  is of type  $c_0\ell_1$  over  $Q$ . Immediately after the corollary, on printed page 27 (PDF page 27), the authors ask:

“We do not know if the converse of Corollary 4.10 holds.”

Figure 1 reproduces the complete source context.

## 2 Definitions

For an arbitrary set  $I$ , the space  $c_0(I)$  consists of the scalar families  $(x_i)_{i \in I}$  such that

$$\{i \in I : |x_i| \geq \varepsilon\} \text{ is finite for every } \varepsilon > 0.$$

The space  $\text{NBV}(K)$  is the source’s right-continuous bounded-variation model of  $C(K)^*$ .

A family  $(F_i)_{i \in I}$  is *of type  $c_0\ell_1$  over  $Q$*  if there are scalar arrays  $(a_{i,t})$  and  $(b_{i,t})$  such that

$$F_i(t) = a_{i,t} + b_{i,t}, \quad (a_{i,t})_{i \in I} \in c_0(I) \quad (t \in Q), \quad \sup_{i \in I} \sum_{t \in Q} |b_{i,t}| < \infty.$$

A family  $(V_t)_{t \in E}$  is *countably separating* for a subspace  $E \subseteq K$  if each  $V_t$  is a neighborhood of  $t$  in  $E$  and every  $M \subseteq E$  satisfying

$$x \in V_y \quad (x, y \in M)$$

is countable.

We use the following finiteness lemma proved by Ronchim and Tausk immediately after their open question.

this example is quite large, namely  $\beth_\omega$ . Recall that  $\beth_\omega = \sup_{n \in \omega} \beth_n$ , where  $\beth_0 = \omega$  and  $\beth_{n+1} = 2^{\beth_n}$  for all  $n \in \omega$ .

**4.1. Countably separating families and families of type  $c_0\ell_1$ .** We show now that the relation that appeared in Lemma 4.4 between spaces admitting countably separating families and weak\*-null families of type  $c_0\ell_1$  is connected to a deeper phenomenon. In this subsection  $K$  denotes again an arbitrary compact line.

**Theorem 4.9.** *Let  $K$  be a compact line and  $Q$  be a subset of  $K$ . If  $Q$  endowed with the topology induced by  $K$  admits a countably separating family then every weak\*-null family  $(F_i)_{i \in I}$  in  $\text{NBV}(K)$  is of type  $c_0\ell_1$  over  $Q$ .*

The following is an immediate consequence of Theorem 4.9

**Corollary 4.10.** *Let  $K$  be a compact line,  $Q$  be a subset of  $K$  and  $(F_i)_{i \in I}$  be a weak\*-null family in  $\text{NBV}(K)$ . If the set  $\{t \in Q : (F_i(t))_{i \in I} \notin c_0(I)\}$  endowed with the topology induced by  $K$  admits a countably separating family then  $(F_i)_{i \in I}$  is of type  $c_0\ell_1$  over  $Q$ .  $\square$*

We do not know if the converse of Corollary 4.10 holds. A proof of such converse would yield a nice topological characterization of weak\*-null families of type  $c_0\ell_1$  and hence (by Theorem 2.7) of  $c_0(I)$ -valued bounded operators defined on the subalgebra induced by a continuous increasing surjection  $\phi : K \rightarrow L$  that admit a bounded extension to  $C(K)$ . This might also lead to interesting characterizations of continuous increasing surjections with the  $c_0(I)$ -extension property, generalizing the main result of [6].

For the proof of Theorem 4.9 we need two lemmas.

**Lemma 4.11** *Let  $K$  be a compact line.  $(F_i)_{i \in I}$  be a weak\*-null family in*

Figure 1: Source Corollary 4.10 and the converse question on printed page 27 of [1].

**Lemma 1** (source Lemma 4.11). *Let  $(F_i)_{i \in I}$  be weak-star-null in  $\text{NBV}(K)$ , let  $J \subseteq K$  be infinite, and let  $\delta > 0$ . Then*

$$\{i \in I : |F_i(t)| \geq \delta \text{ for every } t \in J\}$$

*is finite.*

### 3 Proof intuition

The  $c_0$  part of a  $c_0\ell_1$  decomposition cannot account for an exceptional point, so at each exceptional  $t$  one can select a coordinate  $i(t)$  from the  $\ell_1$  part at which  $F_{i(t)}(t)$  is bounded away from zero. Each fixed coordinate is selected only countably often because an absolutely summable row has countable support. Right continuity keeps the selected function large immediately to the right of  $t$ .

Now suppose a set  $M$  lies mutually in all the chosen neighborhoods. For a fixed threshold,  $F_{i(y)}$  is large on the entire right tail of  $y$  in  $M$ . Source Lemma 4.11 says that an infinite tail allows only finitely many such coordinates. As  $y$  moves to the right, the tails shrink, so these finite coordinate sets form an inclusion chain. A chain of finite sets has countable union. Countable selected-coordinate fibers then force the points with infinite right tail to be countable; the points with finite right tail are automatically countable. This is exactly the countable-separation condition.

## 4 The converse and the exact characterization

**Theorem 2.** *Let  $K$  be a compact line, let  $Q \subseteq K$ , and let  $(F_i)_{i \in I}$  be a weak-star-null family in  $\text{NBV}(K)$ . If  $(F_i)_{i \in I}$  is of type  $c_0\ell_1$  over  $Q$ , then*

$$E = \{t \in Q : (F_i(t))_{i \in I} \notin c_0(I)\}$$

*admits a countably separating family. Consequently,*

$$(F_i)_{i \in I} \text{ is of type } c_0\ell_1 \text{ over } Q \iff E \text{ admits a countably separating family.}$$

*Proof.* Assume that  $(F_i)_{i \in I}$  is of type  $c_0\ell_1$  over  $Q$ , and fix a decomposition

$$F_i(t) = a_{i,t} + b_{i,t} \tag{1}$$

as in the definition. For each  $i \in I$ , the support

$$B_i := \{t \in Q : b_{i,t} \neq 0\}$$

is countable, since  $\sum_{t \in Q} |b_{i,t}| < \infty$ .

Fix  $t \in E$ . Since  $(F_i(t))_{i \in I} \notin c_0(I)$ , there is an integer  $m(t) \geq 1$  for which

$$\{i \in I : |F_i(t)| \geq 1/m(t)\}$$

is infinite. On the other hand,  $(a_{i,t})_{i \in I} \in c_0(I)$ , so all but finitely many indices in this infinite set satisfy  $|a_{i,t}| < 1/(2m(t))$ . Choose one such index and call it  $i(t)$ . By (1),

$$|b_{i(t),t}| \geq \frac{1}{2m(t)} > 0.$$

In particular,

$$\{t \in E : i(t) = i\} \subseteq B_i \tag{2}$$

is countable for every  $i \in I$ .

No point of  $E$  is right-isolated in  $K$ . Indeed, if  $t$  is right-isolated, then  $[\min K, t]$  is clopen, and weak-star nullity applied to its continuous characteristic function gives  $F_i(t) \rightarrow 0$  in the  $c_0(I)$  sense.

Thus right continuity of  $F_{i(t)}$  at  $t$  gives an open convex neighborhood  $U_t$  of  $t$  in  $K$  such that

$$|F_{i(t)}(x)| \geq \frac{1}{2m(t)} \quad \text{whenever } x \in U_t \text{ and } x \geq t. \tag{3}$$

Set  $V_t = U_t \cap E$ . We prove that  $(V_t)_{t \in E}$  is countably separating.

Let  $M \subseteq E$  satisfy  $x \in V_y$  for all  $x, y \in M$ . Partition

$$M = \bigcup_{m \geq 1} M_m, \quad M_m := \{t \in M : m(t) = m\}.$$

Fix  $m$ . For  $y \in M_m$ , define its right tail in  $M_m$  by

$$J_y := \{x \in M_m : x \geq y\}.$$

Since  $M_m \subseteq M \subseteq V_y$  and  $m(y) = m$ , equation (3) shows that

$$|F_{i(y)}(x)| \geq \frac{1}{2m} \quad (x \in J_y). \tag{4}$$

For each  $y$  for which  $J_y$  is infinite, put

$$S_y := \left\{ i \in I : |F_i(x)| \geq \frac{1}{2m} \text{ for every } x \in J_y \right\}.$$

Lemma 1 makes  $S_y$  finite, and (4) gives  $i(y) \in S_y$ . If  $y \leq z$ , then  $J_z \subseteq J_y$ , hence  $S_y \subseteq S_z$ . Therefore the distinct sets  $S_y$  form a chain of finite sets under inclusion. Such a chain has at most one distinct member of each finite cardinality, so

$$S := \bigcup_{\substack{y \in M_m \\ J_y \text{ infinite}}} S_y$$

is countable. By (2), the set of all  $y \in M_m$  with  $J_y$  infinite is contained in a countable union of countable fibers and is therefore countable.

The set of  $y \in M_m$  with  $J_y$  finite is also countable. In fact, if  $y < z$  and both tails are finite, then  $J_z$  is a proper subset of  $J_y$ ; thus the map  $y \mapsto |J_y|$  is injective on these points and takes values in the positive integers. Hence  $M_m$  is countable. Since  $M = \bigcup_{m \geq 1} M_m$ , the set  $M$  is countable.

This proves that  $(V_t)_{t \in E}$  is countably separating. The reverse implication in the final equivalence is precisely source Corollary 4.10.  $\square$

## 5 Verification notes and limitations

1. **One-sided continuity is sufficient.** The construction does not claim that  $F_{i(t)}$  is large to the left of  $t$ . In a clique, only the right tail  $J_t$  is used.
2. **Every exceptional point is eligible.** Right-isolated points cannot belong to  $E$ , so right continuity always supplies the required nontrivial right neighborhood.
3. **No cardinality assumption on  $I$ .** Countability enters only through the support of each absolutely summable row  $b_i$  and the countable threshold partition indexed by  $m$ .
4. **The tail cases are exhaustive.** Infinite tails are controlled by Lemma 4.11 and the nested finite-coordinate sets. Finite tails are countable because their finite cardinalities strictly decrease along the order.
5. **Operator consequence.** Together with source Theorem 2.7, the result gives the topological characterization anticipated after Corollary 4.10 for extendible  $c_0(I)$ -valued operators in the compact-line setting. It does not address extension problems outside that setting.
6. **Computation.** No numerical or finite computation is used or needed; the verification is entirely deductive.

## 6 Bounded novelty check and review recommendation

The run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2111.10670, the exact title, and the core terms. Targeted web searches through 9 August 2026 used the exact sentence “We do not know if the converse of Corollary 4.10 holds”, “countably separating family”, “ $c_0\ell_1$ ”, “ $c_0(I)$ -extension property”, the DOI 10.4064/sm211120-2-6, and the authors’ names. Title/citation searches also checked the later compact-line extension papers returned by the search. No later

source explicitly answering this converse was found. This is a bounded search, not a guarantee of novelty.

**Human-review recommendation: likely valid candidate full solution.** The most useful checks are (i) the exclusion of right-isolated points from  $E$ , (ii) the direction  $J_z \subseteq J_y \Rightarrow S_y \subseteq S_z$ , and (iii) the countable-fiber conclusion from the  $\ell_1$  rows. Once these are confirmed, the proof answers the source question in full.

## References

- [1] V. dos Santos Ronchim and D. V. Tausk, *Extension of  $c_0(I)$ -valued operators on spaces of continuous functions on compact lines*, Studia Math. **268** (2023), 259–289; arXiv:2111.10670. In particular Definition 2.6, Corollary 4.10, and Lemma 4.11.